

Note: $[A + B]^T = A^T + B^T$

- iii) $A + \mathbf{0} = \mathbf{0} + A = A$, where $\mathbf{0}$ denotes zero matrix of the same order. Thus zero matrix plays the same role in matrix algebra as zero in ordinary algebra.

Some Results:

- a) $A + (-A) = (-A) + A = \mathbf{0}$
- b) $A + A = 2A$
- c) $k(A + B) = kA + kB$, for any constant k
- d) $(k_1 + k_2)A = k_1A + k_2A$, for any constants k_1 and k_2 .

5.3.3.3 Matrix Multiplication

If the number of columns of a matrix A be equal to the number of rows of another matrix B , then the matrices A and B are said to be '*conformable for the product*' AB , and the product AB is said to be defined.

If $A = [a_{ij}]_{m \times n}$ and $B = [b_{jk}]_{n \times p}$,

then $AB = [c_{ik}]_{m \times p}$ where $c_{ik} = \sum_{j=1}^n a_{ij} b_{jk}$

$$\text{Thus, } \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 3 \\ 5 & 2 \\ 9 & 7 \end{bmatrix} = \begin{bmatrix} 1+10+27 & 3+4+21 \\ 0+5+18 & 0+2+14 \end{bmatrix} = \begin{bmatrix} 38 & 28 \\ 23 & 16 \end{bmatrix}$$

Some Properties:

- i) In general, $AB \neq BA$ [so, matrix multiplication is not commutative in general.]
- ii) For conformable matrices $(AB)C = A(BC)$ [so, matrix multiplication is associative.]
- iii) $(B+C)D = BD + CD$ [so, matrix multiplication is distributive with respect to addition]
- iv) $(AB)^T = B^T \cdot A^T$

5.3.3.4 Adjoint and Reciprocal Matrices

If $A = [a_{ij}]_{n \times n}$ be a square matrix, then the transpose of the matrix $[A_{ij}]_{n \times n}$ whose elements are the cofactors of the corresponding element in $|A|$ is called the 'adjoint' or 'adjugate' matrix of A and is denoted by **adjA**. It is equal to $[A_{ji}]_{n \times n}$.

5.3.3.5 Trace of a Matrix

The trace of a square matrix is the sum of its diagonal elements.

Thus, in the matrix $\begin{bmatrix} 1 & 1 & 2 \\ 2 & 0 & 1 \\ 3 & 3 & -1 \end{bmatrix}$ the trace (tr) is $= 1 + 0 - 1 = 0$

In general, trace $\text{tr}(A) = a_{11} + a_{22} + \dots + a_{nn} = n \sum_{i=1}^n a_{ii}$

- Properties:**
- i) $\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$
 - ii) $\text{tr}(A') = \text{tr}(A)$
 - iii) $\text{tr}(AB) = \text{tr}(BA)$ and
 - iv) $\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$.

5.3.3.6 Sub Matrices and Minors

Any matrix obtained by omitting some rows and some columns of a given $m \times n$ matrix A is called 'sub matrix' of A .

Thus, $\begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$ is a sub matrix of $\begin{bmatrix} a_1 & b_1 & c_1 & d_1 \\ a_2 & b_2 & c_2 & d_2 \\ a_3 & b_3 & c_3 & d_3 \end{bmatrix}$

The determinant of the square matrix of order ' r ', obtained from a given $m \times n$ matrix A by omitting $(m - r)$ rows and $(n - r)$ columns, is called a '**minor**' of A of order ' r '.

Thus, $\begin{bmatrix} a & h \\ h & b \end{bmatrix}$ is a minor of $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$ of order 2

5.3.3.7 Transpose of Sum or Difference of Matrices

The transpose of a sum or difference of matrices is equal to the sum or difference of the transpose of the matrices. Thus,

$$(A_{m \times n} \pm B_{m \times n} \pm C_{m \times n})' = A'_{m \times n} \pm B'_{m \times n} \pm C'_{m \times n}$$

Example: $A = \begin{bmatrix} 2 & -1 \\ 0 & -1 \end{bmatrix}$ $B = \begin{bmatrix} 6 & 8 \\ 10 & -12 \end{bmatrix}$ $C = \begin{bmatrix} 0 & -1 \\ 0 & 0 \end{bmatrix}$ then

$$[A + B + C]' = \begin{bmatrix} -4 & -10 \\ -10 & 11 \end{bmatrix}' = \begin{bmatrix} -4 & -10 \\ -10 & 11 \end{bmatrix}$$

or alternatively

$$A' - B' + C' = \begin{bmatrix} 2 & 0 \\ -1 & -1 \end{bmatrix} - \begin{bmatrix} 6 & 10 \\ 8 & -12 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ -1 & 0 \end{bmatrix} = \begin{bmatrix} -4 & -10 \\ -10 & 11 \end{bmatrix}$$

5.3.3.8 Transpose of a Product of Matrices

The transpose of a product of matrices is equal to the product of the transpose of the matrices in inverse sequence; thus

$$\begin{bmatrix} A_{m \times n} & B_{n \times p} & C_{p \times q} \end{bmatrix}' = C_{q \times p}' B_{p \times n}' A_{n \times m}'$$

Example: $A = \begin{bmatrix} 3 & 0 \\ -4 & -1 \end{bmatrix}$ $B = \begin{bmatrix} 3 & 5 & -7 \\ 0 & -1 & 8 \end{bmatrix}$ $C = \begin{bmatrix} 6 \\ -1 \\ 0 \end{bmatrix}$ then

$$\begin{aligned} ABC &= \begin{bmatrix} 39 \\ -53 \end{bmatrix} \quad [ABC]' = [39 \quad -53] = C'B'A' \\ &= [6 \quad -1 \quad 0] \begin{bmatrix} 3 & 0 \\ 5 & -1 \\ -7 & 8 \end{bmatrix} \begin{bmatrix} 3 & -4 \\ 0 & -1 \end{bmatrix} \end{aligned}$$

5.3.4 Matrix Inversion

A square matrix $A_{n \times n}$ with $|A| \neq 0$ is called a **non-singular** matrix. $A_{n \times n}$ is called **singular** matrix if $|A| = 0$.

Definition: Inverse/reciprocal of a non-singular $n \times n$ square matrix A is a matrix B of the same order as that A , which satisfies the matrix equation $AB = BA = I$, where I is the $n \times n$ unit matrix. Inverse/reciprocal of matrix A is denoted as A^{-1} .

It can be shown that the *inverse* of a non-singular matrix A is given by

$$A^{-1} = \frac{1}{|A|} \cdot \text{Adj.} A$$

Example: $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & -1 & 3 \\ 3 & 2 & -1 \end{bmatrix}$ $\therefore |A| = 1(1-6) - 1(-2-9) + 1(4+3)$
 $= -5 + 11 + 7 = 13$ i.e., $|A| \neq 0$

So A is a non-singular matrix, its inverse exists.

Now, matrix of cofactors of $A = \begin{bmatrix} -5 & 11 & 7 \\ 3 & -4 & 1 \\ 4 & -2 & -3 \end{bmatrix}$

$$\therefore \text{Adj.} A = \begin{bmatrix} -5 & 3 & 4 \\ 11 & -4 & 1 \\ 7 & 1 & -3 \end{bmatrix} \quad \text{and} \quad A^{-1} = \frac{1}{13} \begin{bmatrix} -5 & 3 & 4 \\ 11 & -4 & 1 \\ 7 & 1 & -3 \end{bmatrix}$$

Some Properties:

- The inverse of a matrix, if exists, is unique.
- The necessary and sufficient condition that the inverse of a square matrix A exist, is that A is non-singular.
- $[AB]^{-1} = B^{-1} \cdot A^{-1}$ (this is known as 'law of reversal of inverse')

iv) $(A^{-1})^{-1} = A$

v) $(A^{-1})^T = (A^T)^{-1}$

and $(A^{-1})^T \cdot A^T = I = A^T (A^{-1})^T$ (operations of reciprocating and transposing are commutative)

Check Your Progress 3

1) Obtain the inverse of $A = \begin{vmatrix} 1 & 2 & 3 \\ 1 & 3 & 5 \\ 1 & 5 & 12 \end{vmatrix}$

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2) Prove that, if $A = \begin{bmatrix} 2 & -3 & 0 \\ 3 & 1 & -2 \\ -1 & 0 & -4 \end{bmatrix}$ then $A^{-1} = -\frac{1}{50} \begin{bmatrix} -4 & -12 & 6 \\ 14 & -8 & 4 \\ 1 & 3 & 11 \end{bmatrix}$

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5.3.5 Some Other Types of Matrices

5.3.5.1 Orthogonal Matrix

A square matrix A is said to be orthogonal, if its transpose A^T produces a unit matrix when multiplied by A , i.e., $A^T \cdot A = I$.

Some Properties of orthogonal matrices:

i) Orthogonal matrices are non-singular,
 i.e., if $A^T \cdot A = I$ then $|A| \neq 0$

ii) Unit matrices are orthogonal, i.e.,
 $I^T \cdot I = I$ $I = I$

iii) The value of the determinant of an orthogonal matrix is either 1 or (-1). If A is an orthogonal matrix, then $A^T A = I$
 $\therefore |A^T \cdot A| = |A^T| \cdot |A| = |A|^2 = |I|$
 $\therefore |A| = \pm 1$

iv) The product of two orthogonal matrices is orthogonal.

v) The transpose of an orthogonal matrix is orthogonal.

vi) The inverse of an orthogonal matrix is orthogonal.

5.3.5.2 Symmetric and Skew-Symmetric Matrices

A square matrix A is said to be 'symmetric' if its transpose be equal to it, i.e., if $A^T = A$. Hence, the matrix $[a_{ij}]$ is symmetric, if $a_{ij} = a_{ji}$ for all i and j . For example,

$$\begin{bmatrix} a & b & c \\ b & e & d \\ c & d & f \end{bmatrix} \text{ is a symmetric matrix.}$$

A square matrix A is said to be 'skew-symmetric', if its transpose is equal to its negative. i.e., if $A^T = -A$. Hence, the matrix $[a_{ij}]$ is skew-symmetric, if $a_{ij} = -a_{ji}$ for all i and j . The diagonal elements of skew – symmetric matrix are obviously zero. For example,

$$\begin{bmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{bmatrix} \text{ is a skew – symmetric matrix.}$$

Some Properties:

- i) The product of a matrix and its transpose is a symmetric matrix

$$\text{Now, } [A A^T]^T = [A^T]^T \cdot [A]^T = A \cdot [A]^T$$

Hence, $A \cdot A^T$ is symmetric.

- ii) The product of two symmetric matrices of the same order is symmetric if the product is commutative.

If A and B be two symmetric matrices of the same order and $AB = BA$, then

$$[AB]^T = B^T \cdot A^T = B \cdot A = AB.$$

Hence, AB is symmetric.

- iii) If A is a square matrix, then $(A + A^T)$ is symmetric and $(A - A^T)$ is skew-symmetric.

$$(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$$

Thus, $(A + A^T)$ is symmetric.

$$\text{Similarly, } (A - A^T)^T = A^T - (A^T)^T = A^T - A = -(A - A^T)$$

Hence, $(A - A^T)$ is skew-symmetric.

- iv) Every square matrix can be expressed as a sum of a symmetric matrix and a skew symmetric matrix uniquely.

Let A be a square matrix, then,

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

5.3.5.3 Idempotent Matrix

A symmetric matrix that reproduces itself when multiplied by itself is termed as an idempotent matrix.

That is, A will be termed as idempotent matrix if $AA = A$.

It is easily observed that for an idempotent matrix A , $A^n = A$, for all natural numbers n .

5.3.6 Partitioned Matrices

It is often convenient to partition a matrix into sub matrices. These sub matrices can then be treated as scalars (single values) in performing operations on the original matrix. Partitioning of a matrix is indicated by horizontal and vertical lines between rows and columns.

The $m \times n$ matrix A may be partitioned as follows $A = (A_1 : A_2)$

If they are partitioned conformably, partitioned matrices can be added, subtracted or multiplied. If an $m \times n$ matrix A is partitioned $A = (A_1 : A_2)$, where A_1 is $m \times n_1$, A_2 is $m \times n_2$ and $n_1 + n_2 = n$, and $m \times n$ matrix B is partitioned $B = (B_1 : B_2)$, where B_1 is $m \times n_1$, B_2 is $m \times n_2$ and $n_1 + n_2 = n$, then $A \pm B = [A_1 \pm B_1 : A_2 \pm B_2]$

Similarly, if $A = \begin{bmatrix} A_1 \\ \vdots \\ A_2 \end{bmatrix}$, where A is $m \times n$, A_1 is $m_1 \times n$, A_2 is $m_2 \times n$ and

$m_1 + m_2 = m$, and $B = \begin{bmatrix} B_1 \\ \vdots \\ B_2 \end{bmatrix}$, where B is $m \times n$, B_1 is $m_1 \times n$, B_2 is $m_2 \times n$ and

$m_1 + m_2 = m$ then $A \pm B = \begin{bmatrix} A_1 \pm B_1 \\ \vdots \\ A_2 \pm B_2 \end{bmatrix}$

Partitioning is frequently conceptually convenient when matrices are to be added or subtracted, but the computational advantages of matrix partitioning are associated primarily with multiplication and other more complex operations. Partitioned matrices must be partitioned conformable for multiplication. If matrix A of order $m \times n$ is partitioned $A = (A_1 : A_2)$, where A_1 is $m \times n_1$, A_2 is $m \times n_2$ and

$n_1 + n_2 = n$ and $n \times p$ matrix B is partitioned $B = \begin{bmatrix} B_1 \\ \vdots \\ B_2 \end{bmatrix}$, where B_1 is $n_1 \times p$ and B_2 is

$n_2 \times p$ then $AB = [A_1 : A_2] \begin{bmatrix} B_1 \\ \vdots \\ B_2 \end{bmatrix} = A_1 B_1 + A_2 B_2$

Example: if $A = \begin{bmatrix} A_{11} & \vdots & A_{12} \\ \vdots & \ddots & \vdots \\ A_{21} & \vdots & A_{22} \end{bmatrix}$ where A_{11} is $m_1 \times n_1$, A_{12} is $m_1 \times n_2$, A_{21} is

$m_2 \times n_1$, A_{22} is $m_2 \times n_2$ and $m_1 + m_2 = m$, $n_1 + n_2 = n$ and if

$$B = \begin{bmatrix} B_{11} & \vdots & B_{12} \\ \dots & \dots & \dots \\ B_{21} & \vdots & B_{22} \end{bmatrix}, \text{ where } B_{11} \text{ is } n_1 \times p_1, B_{12} \text{ is } n_1 \times p_2, B_{21} \text{ is } n_2 \times p_1, B_{22} \text{ is } n_2 \times p_2$$

and $n_1 + n_2 = n$, $p_1 + p_2 = p$, then A and B are partitioned conformably for

$$\begin{aligned} \text{multiplication and } AB &= \begin{bmatrix} A_{11} & \vdots & A_{12} \\ \dots & \dots & \dots \\ A_{21} & \vdots & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & \vdots & B_{12} \\ \dots & \dots & \dots \\ B_{21} & \vdots & B_{22} \end{bmatrix} \\ &= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & \vdots & A_{11}B_{12} + A_{12}B_{22} \\ \dots & \dots & \dots \\ A_{21}B_{11} + A_{22}B_{21} & \vdots & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix} \end{aligned}$$

Check Your Progress 4

- 1) List the properties of orthogonal matrix.

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- 2) Define an idempotent matrix.

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- 3) If

$$A = \begin{pmatrix} 1 & 3 & 1 \\ -1 & 0 & 1 \\ 2 & -1 & 4 \\ 0 & 2 & -3 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & -2 & 1 & 0 & -1 \\ 5 & -1 & 4 & -3 & 2 \\ 3 & -2 & 0 & 1 & -1 \end{pmatrix}$$

$$\text{then show } AB = \begin{bmatrix} A_{11} & \vdots & A_{12} \\ \dots & \dots & \dots \\ A_{21} & \vdots & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & \vdots & B_{12} \\ \dots & \dots & \dots \\ B_{21} & \vdots & B_{22} \end{bmatrix}$$

$$= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & \vdots & A_{11}B_{12} + A_{12}B_{22} \\ \dots & \dots & \dots \\ A_{21}B_{11} + A_{22}B_{21} & \vdots & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}.$$

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5.3.7 Rank of a Matrix

A number 'r' is said to be the rank of a matrix A, if

- i) there is at least one (r x r) sub-matrix of A whose determinant is not equal to zero; and
- ii) the determinant of every (r+1)-rowed square sub-matrix of A is zero.

In other words, rank of a matrix is equal to the order of the highest order non-singular square matrix contained in A.

For example, the rank of matrix $\begin{bmatrix} 5 & 2 \\ -2 & -3 \end{bmatrix}$ is 2; because this second order matrix corresponds to the determinant $\begin{bmatrix} 5 & 2 \\ -2 & -3 \end{bmatrix}$, which is not zero, hence this given 2 x 2 matrix contains a non-singular matrix of order 2.

Whereas, the rank of the matrix $\begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$ is 1; the highest order determinant $\begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$ is zero, next to this would be of the 1st order, which is non-zero.

Again, the rank of $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ is zero.

Note:

- i) The rank of a matrix cannot exceed the number of its rows or columns - whichever is less.
- ii) The rank of a column or row matrix is at most 1, unless the matrix is a null matrix, where the rank is always zero.
- iii) Rank of the transpose of a matrix is the same as the rank of that matrix.

Example: Determine the rank of the matrix $\begin{bmatrix} 2 & -1 & 3 & 1 & 5 \\ 3 & 2 & 0 & 2 & 1 \end{bmatrix}$. Here, picking any sub-matrix of order 2, it can easily be shown that value of the determinant is non-zero, hence rank = 2.

The process of computing rank of a matrix may be simplified by applying row/column operations which do not change the values of the corresponding determinant, as discussed earlier.

$$\begin{bmatrix} 2 & -1 & 3 & 1 & 5 \\ 3 & 2 & 0 & 2 & 1 \end{bmatrix} \sim \begin{bmatrix} 0 & 0 & 0 & 1 & 5 \\ -1 & 4 & -6 & 2 & 1 \end{bmatrix}$$

by,

$$c_1' \rightarrow c_1 - 2c_4$$

$$c_2' \rightarrow c_2 + c_4$$

$$c_2' \rightarrow c_2 - c_4$$

$$\sim \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ -1 & 4 & -6 & 2 & -9 \end{bmatrix}, \text{ by, } c_5' \rightarrow c_5 - 5c_4$$

$$\sim \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 & -9 \end{bmatrix}, \text{ by, } \begin{aligned} c_1'' &\rightarrow 6c_1' - c_3' \\ c_2'' &\rightarrow 6c_2' - 4c_3' \\ c_3'' &\rightarrow 9c_3' - 6c_5' \end{aligned}$$

$$\sim \begin{bmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \text{ by, } R_2' \Rightarrow R_2 - 2R_1$$

$$\sim [0 \ I_2], \text{ by, } R_2'' \Rightarrow \frac{1}{9}R_2'$$

So, rank of the matrix is 2

Check Your Progress 5

- 1) Show that rank of $A=2$ and rank of $B=2$, where

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 2 & -3 & 4 \\ 3 & -2 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} -1 & -2 & 3 \\ 6 & 12 & 6 \\ 5 & 10 & 5 \end{bmatrix}$$

but rank of $(A+B)=3$.

Show further that, rank of $(AB) = 1$, while rank of $(BA) = 2$.

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- 2) How would you determine linear dependence of matrix?

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- 3) Define the rank of matrix in terms of linear in dependence.

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5.4 LET US SUM UP

In this unit, we explained the concept of determinant, and the related concepts of minors, cofactors, adjoint and reciprocal determinants, symmetric determinant, skew ad skew-symmetric determinants. Next, we discussed the properties of determinants; how to find product of two determinants. Further, we discussed the process of solving a system of simultaneous linear equations by Cramer's rule based on determinants.

Then, we discussed another related concept that of matrix, its types, and operations on matrices. We explained and used the related concepts of submatrices and minors, orthogonal matrix, symmetric and skew-symmetric matrices, idempotent matrix, adjoint and reciprocal matrices, trace of a matrix, transpose of a matrix, non-singular matrix, inverse of a matrix. We explained the methods of finding (i) inverse of a non-singular matrix, (ii) the rank of a matrix. Finally, we discussed how to partition matrices, and use these in carrying out efficiently matrix operations.

5.5 KEY WORDS

Cofactor : The 'cofactor' of any element in a determinant is its coefficient in the expansion of the determinant. It is therefore equal to the corresponding minor with a proper sign. The sign of the cofactor of an element in the i^{th} row and j^{th} column is $(-1)^{i+j}$.

Determinants : If the linear equation $a_1x + b_1 = 0$ and $a_2x + b_2 = 0$ have the same solution, then $\frac{a_1}{a_2} = \frac{b_1}{b_2}$, i.e., $\frac{a_1}{b_1} = \frac{a_2}{b_2}$, or, $a_1b_2 - a_2b_1 = 0$. The expression $(a_1b_2 - a_2b_1)$ is called a determinant and is denoted by the symbol $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$.

Inverse of a Matrix : A square matrix $a_{n \times n}$ with $|A| \neq 0$ is called a non-singular matrix. The inverse of a non-singular matrix A is given by $A^{-1} = \frac{1}{|A|} \cdot \text{Adj.}A$.

Matrix : A 'matrix' is defined as a rectangular array of elements arranged in rows and columns and 'matrix algebra' basically provides methods for dealing with sets of simultaneous relationships involving many